1. What is Python?
2. Who developed Python and in which year?
3. Why is Python called a high-level language?
4. Why is Python considered platform-independent?
5. What are the features of Python?
6. Why is Python dynamically typed?
7. What does it mean that Python is an interpreted language?
8. What are Python applications in real life?
9. What is the difference between Python 2 and Python 3?
10. Why should a beginner choose Python?
11. What is a variable?
12. How does Python allocate memory to variables?
13. What is the id() function?
14. What is Multiple Variable Creation (MVC)?
15. What are keywords?
16. Why can't we use keywords as identifiers?
17. What are identifiers?
18. What are the rules for naming identifiers?
19. What are literals in Python?
20. What is the difference between variable and constant?
21. What are Python datatypes?
22. What is the int datatype?
23. What is the float datatype?
24. What is a complex number?
25. What is the bool datatype?
26. What is a string?
27. What does immutability mean in strings?
28. What is indexing?
29. What is slicing?
30. What is a list?
31. Why are lists mutable?
32. What is a tuple?
33. How is a tuple different from a list?
34. What is a set?
35. What is a dictionary?
36. Why does Python support 3 types of quotes?
37. What are string functions?
38. What is the difference between find() and index()?
39. What does split() do?
40. What does join() do?
41. What is string slicing?

42. What is negative indexing?
43. What is ASCII value?
44. How do you convert string to uppercase or lowercase?
45. What is replace() used for?
46. What are list methods?
47. What does append() do?
48. What is difference between append() and extend()?
49. How to remove elements from a list?
50. What is list comprehension?
51. What are tuple packing and unpacking?
52. Why are tuples faster than lists?
53. What are set properties?
54. Why are sets unordered?
55. What is a frozenset?
56. Can sets store duplicate values?
57. What is a dictionary?
58. What are dictionary keys?
59. Why must keys be unique?
60. What is the difference between get() and direct indexing?
61. What is items()?
62. What is values()?
63. What is hashing in Python?
64. What is the difference between list and dictionary?
65. What are nested collections?
66. What is type casting?
67. What is explicit type casting?
68. What is implicit type casting?
69. Can we convert string to list?
70. What is the difference between str() and repr()?
71. What happens when converting float to int?
72. Can we convert complex to integer?
73. What functions are used for type conversion?
74. What is eval()?
75. Why is eval() dangerous?
76. What is reference copy?
77. What is shallow copy?
78. Which module is used for shallow copy?
79. What is deep copy?
80. How does deep copy work?
81. Why is copy important in lists?
82. What is the difference between copy() and slicing?

83. Does shallow copy duplicate nested lists?
84. When is deep copy required?
85. What is mutability?
86. What are arithmetic operators?
87. Difference between / and //?
88. What is modulus operator?
89. What is exponentiation operator?
90. What are relational operators?
91. What are logical operators?
92. What is short-circuit evaluation?
93. What are identity operators?
94. Difference between is and ==?
95. What are membership operators?
96. What are bitwise operators?
97. What does ~ do?
98. What is left shift operator?
99. What is right shift operator?
100. What is operator precedence?
101. What is the input() function?
102. Why does input() return string?
103. What is type-casting with input?
104. What is the print function?
105. What are print formatting methods?
106. What is conditional branching?
107. What is an if statement?
108. What is nested if?
109. Why do we use elif?
110. What are loops?
111. What is the difference between for and while?
112. What is range()?
113. What is break statement?
114. What is continue?
115. What is pass?
116. What is pattern printing?
117. Why do companies ask pattern logic?
118. How do nested loops work in patterns?
119. What is star pattern?
120. What is number pattern?
121. What is an alphabet pattern?
122. How to print diagonals?
123. How to print hollow patterns?

124. Why is pattern logic important?
125. What is time complexity of pattern loops?
126. What is a function?
127. Why do we need functions?
128. What are the 4 types of functions?
129. What is return statement?
130. What are local variables?
131. What are global variables?
132. What is global keyword?
133. What are positional arguments?
134. What are keyword arguments?
135. What are default arguments?
136. What are variable-length arguments?
137. What is *args?
138. What is **kwargs?
139. What is recursion?
140. What is base condition?
141. What is OOP?
142. What is a class?
143. What is an object?
144. What is constructor?
145. What is __init__?
146. What is method?
147. Difference between method and function?
148. What is class variable?
149. What is instance variable?
150. What is static method?
151. What is inheritance?
152. What is single inheritance?
153. What is multiple inheritance?
154. What is multi-level inheritance?
155. What is hierarchical inheritance?
156. What is hybrid inheritance?
157. What is constructor chaining?
158. What is method overriding?
159. What is polymorphism?
160. What is operator overloading?
161. What is encapsulation?
162. What are access specifiers?
163. How to access private variables?
164. What is abstraction?

165. What is abstract class?
166. What is lambda function?
167. Where do we use lambda?
168. What is map()?
169. What is filter()?
170. What is reduce()?
171. Can lambda contain multiple lines?
172. What is anonymous function?
173. What is difference between map and filter?
174. What is list mapping?
175. What is function mapping?
176. What is list comprehension?
177. What is set comprehension?
178. What is dictionary comprehension?
179. Why are comprehensions faster?
180. What is conditional comprehension?
181. What is nested comprehension?
182. Can comprehensions replace loops?
183. Why should comprehensions be readable?
184. Can we use if-else in comprehension?
185. What is Cartesian product using comprehension?
186. What is a decorator?
187. What is a generator?
188. What is yield keyword?
189. Difference between generator and function?
190. What is iterator?
191. What is iterator protocol?
192. What is next()?
193. Why are generators memory-efficient?
194. What is chaining decorators?
195. What is practical use of decorators?
196. What is file handling?
197. What are file modes?
198. What is the difference between text and binary files?
199. What does read(), readline(), readlines() do?
200. What is CSV file?
201. What is a database connection in Python?
202. Which Python module is used to connect to SQLite databases?
203. What is the purpose of connect() function in SQLite?
204. What is a cursor object?
205. Why do we use cursor() method?

206. What is the purpose of execute() in SQLite?
207. Why is commit() used in database operations?
208. What happens if we forget to call commit()?
209. What does close() function do in database connection?
210. What is the difference between SQL and SQLite?
211. What are CRUD operations?
212. What is a parameterized query?
213. Why should we avoid string-based SQL queries?
214. What is SQL injection?
215. What is the purpose of ? placeholder in SQLite queries?
216. What is a module in Python?
217. What is a package in Python?
218. What is the difference between package and module?
219. What is the purpose of __init__.py?
220. Can a package contain another package?
221. What are namespace packages?
222. What are the different ways to import a module?
223. What is from module import function syntax?
224. What is the use of aliasing modules?
225. Why is import * not recommended?
226. What is a built-in module?
227. Name some commonly used built-in modules in Python.
228. What is module reloading?
229. Why should large programs be split into modules?
230. What happens if two modules have the same function name?
231. What is an exception?
232. What is exception handling?
233. What is the purpose of try block?
234. What is except block used for?
235. Why should specific exceptions be caught before generic exceptions?
236. What is finally block?
237. When does a finally block execute?
238. What is the purpose of the else block in exception handling?
239. What is a runtime error?
240. What is the difference between error and exception?
241. What is raise keyword?
242. What is user-defined exception?
243. What happens if an exception is not handled?
244. What is multiple exception handling?
245. What is exception chaining?
246. What is JSON?

247. Why is JSON widely used in APIs?
248. What is the Python module for handling JSON?
249. What does json.dumps() do?
250. What does json.loads() do?
251. What are the differences between JSON and dictionary?
252. What is Pickle in Python?
253. Why is Pickle used?
254. What does pickle.dump() do?
255. What does pickle.load() do?
256. What is a regular expression?
257. Why do we use regex in Python?
258. Which Python module is used for regex?
259. Difference between match() and search().
260. What is the use of findall()?
261. What is the use of sub()?
262. What does \d represent?
263. What does \w represent?
264. What does \s represent?
265. Difference between * and +.
266. Difference between {m} and {m,n}.
267. What is a character class?
268. What is a capturing group?
269. What is a non-capturing group?
270. What does ^ do in regex?
271. What does $ do in regex?
272. What is greedy vs non-greedy matching?
273. What is a lookahead?
274. What is a lookbehind?
275. Why is regex used in validations?
276. What is SQL?
277. Difference between SQL and SQLite.
278. What is a table?
279. What is a primary key?
280. What is a foreign key?
281. What is a constraint?
282. What is NOT NULL constraint?
283. What is UNIQUE constraint?
284. What is CHECK constraint?
285. What is normalization?

286. Difference between DELETE, DROP, TRUNCATE.
287. What is a query?
288. What is a cursor in Python SQL connection?
289. What is a parameterized query?
290. What is SQL injection?
291. What is commit() in SQLite?
292. What is indexing?
293. What is aggregation in SQL?
294. What is GROUP BY?
295. What is HAVING?
296. What is SQLite3 in Python?
297. How do we import SQLite in Python?
298. What is the purpose of sqlite3.connect()?
299. What is a cursor in SQLite?
300. Why do we use cursor() method?
301. What does cursor.execute() do?
302. Why is the commit() method necessary?
303. What happens if we forget to call commit()?
304. What is fetchone()?
305. What is fetchall()?
306. Why do we use parameterized queries in Python SQL?
307. What is SQL injection and how to prevent it in Python?
308. What is the role of ? placeholders in SQLite Python queries?
309. What are CRUD operations in SQL?
310. How do we close a database connection in Python?